Spring 2026 — Project

Project overview
Proposal
Milestone 1
Milestone 2
Final deliverable
Showtime - Vote
Special helper EC

Final deliverable

Due 27-April-2026, 1pm – <u>before class</u>

(no late submission, no extension)

Purpose:

- Practice and experience database programming
- Transition from concepts to practice
- Evaluate your project milestone for compliance with the requirements
- Work as a team, collaborate with your project partners professionally and contribute fairly

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Remember to ***back up your project!!***

Final step!

You will work in your project group to develop a software that uses a relational database to provide services.

Minimum requirements

- If you did the extra credit milestone (DB programming), continue and complete the remaining functionalities/features of your project and complete all project requirements
  [03/20: Removed. The extra credit milestone is not available for Spring 2026. It was from Fall 2025 semester.]
- Write programs to interact with the database you designed and set up in previous milestones

- Implement all functionalities and necessary features you proposed to fulfill the project requirements
- Include **dynamic behaviors**, where the front end (user interface) respond to user inputs or actions (or web services) and updates the screen (interface) accordingly
- Allow users to perform the following functionalities. It is important to note that these functionalities must be done **through your software**; users are not allowed to access your database directly. Manually performing any of the following functionalities (e.g., through phpMyAdmin or MySQL/SQL/DBMS command line) does **not** satisfy these requirements.
  - **Retrieve** data from the database
  - **Add** data to the database
  - **Update** data in the database
  - **Delete** data from the database
  - Do at least **one** of the following:

    > **Important note:**
    > If your team consists of 5 members, you are required to have **all four** of following functionalities

    - Sort data
    - Search or filter data
    - Export data (from the database) in some forms (such as JSON, XML, HTML, or CSV)
    - Upload / import data (in some forms such as JSON, XML, HTML, or CSV) into the database
- Support **multiple users**, allowing users to use the app **simultaneously** without interfering with each other.
  - Your database must be shared or hosted on a public server to provide centralized services — allowing your app to **access a single shared database** that can be accessed by multiple users concurrently. [Refer to project proposal page for some ideas on how you may test your app]
    - Setting up the same database locally on multiple machines is not directly considered a "shared database." In fact, they, simply, are duplicate databases that need to be synchronized either manually or using the concepts of distributed database system.
      - If you choose to host your database locally, you would need a workaround when testing your app.
    - To be realistic, try avoid setting up a database that can only be accessed by a single machine and that need to be synchronized manually.
    - To simplify the process and minimize the overhead (in terms of configuration), do not set up a distributed database system.

        - If you would rather explore something new or prefer challenges, you may try to set up a distributed database system. Please spare time for troubleshooting and plan your time very well.
    - If you implement a web-based project
        - A simple way to test this feature is to open multiple different web browsers, then on different web browsers try to access / use the service your app provides concurrently.
    - If you implement a non web-based project
        - Install your app on each user' machine. Host your database on a public server so that it can be shared. To test this feature, try to access / use the service your app provides from multiple machines concurrently.
        - Alternatively, you may set up multiple virtual environments on a machine; each environment represents each user's machine. To test this feature, try to access / use the service your app provides from multiple environments concurrently. Please spare time for troubleshooting and plan your time very well.

- Support **returning users** — allowing returning users to access / retrieve / manipulate their existing data.

- Set up **database security at the database level**
    - This typically involves limited access or privilege on the database system and may differentiate the levels of privileges of
        - Developers (you and your teammates) — which tables can be accessed and what operations an individual developer can perform on the tables
        - Users (end users who use your app) — which tables can be accessed by your app to provide services to an individual user (e.g., consider different categories of users: students vs. instructors; which tables / operations a student user can access / perform, which tables / operations an instructor user can access / perform)

- Incorporate **database security at the application level**
    - This may involve, for example,
        - User authentication to use some (or all) services your app provides
        - Password hashing
        - Control of execution flow such as redirecting a user to a certain portion of your service
        - Providing users with different views or interfaces, limiting access to certain data or services

- Reminder: your project must meet the minimum number of table requirements. All the normalized tables must be set up in the previous milestones and all of them must be utilized in some way by your app.

- [**Extra credits**] **Host your database and app on GCP**
    - Deploy your database and app on GCP

- Submit a URL to your app hosted on GCP with your source code
- Use this URL to demo your app
- Submit screenshot(s) showing your database on GCP

Submit a brief report in five parts

1. Database design
   - Include a final version of your E-R diagram
   - Include a final version of your tables written in schema statements

2. Database programming
   - Specify where you host your database
     - If you do the extra credits (hosting your database and your app on GCP), submit screenshot(s) showing your database on GCP
   - Specify where you host your app — the deployment environment needed to deploy and run your project
     - If you do the extra credits (hosting your database and your app on GCP), submit the URL to your app
   - Describe and include all instructions / steps needed to deploy and run your project
     - Note: we may randomly set up, deploy, and run your project on a similar environment. It is important that you provide the description and a complete set of the instructions.
   - Refer to advanced SQL commands your wrote in the previous milestone, clearly discuss how the advanced SQL commands are incorporated in your app, what features of your app use them, and how they reflect the database

3. Describe the database security at the database level
   - Specify whether the security is set for developers or end users
   - Discuss how you set up security at the database level (access control)
   - Submit the SQL commands you use to limited / set privileges such as

     `GRANT privilege-name ON object-name TO {user-name | PUBLIC | role-name}` or

     `REVOKE privilege-name ON object-name FROM {user-name | PUBLIC | role-name}`

4. Describe the database security at the application level
   - Discuss how database security at the application level is incorporated in your project.
   - Submit code snippet(s) to illustrate how security aspect is implemented and to support your discussion.

5. Reflection and peer evaluation *(due when the final deliverable is due)*
   - **This is an individual task.** Do **not** include it in the project's final report.

- The teaching team will consider the reflection and evaluation when assigning the final deliverable grade to an individual team member. Each team member's grade may be adjusted by 0%-100% deduction, based on his/her contribution.
- Submit your reflection and peer evaluation, ***https://forms.office.com/r/8XCV8P7utY (https://forms.office.com/r/8XCV8P7utY)***
  - Note: everyone is required to submit the reflection and peer evaluation, even though you are working on the project individually.
- You are **required** to enter the names and computingIDs of all team members
- If you work on the project by yourself, indicate in the reflection and peer evaluation form that you work by yourself
- Once this form is closed, the form will not be reopened and we have to assign a zero grade to this section of the rubric.

Grading Rubric

[Total: 200 points + 10 extra points]

(35 points) Brief report:

- (4 points) Database design
  - (2 points) — Include a final version of your E-R diagram
  - (2 points) — Include a final version of your tables written in schema statements

- (10 points) Database programming
  - (1 point) — Specify where you host your database
  - (1 point) — Specify where you host your app — the deployment environment needed to deploy and run your project
  - (2 points) — Describe and include all instructions / steps needed to deploy and run your project
  - (6 points) — Discuss how the advanced SQL commands are incorporated in your app, what features of your app use them, and how they reflect the database

- (8 points) Database security at the database level
  - (5 points) — Discuss how you set up security at the database level (access control). Also specify whether the security is set for developers or end users.
  - (3 points) — Submit the SQL commands you use to limited / set privileges

- (8 points) Database security at the application level
  - (5 points) — Discuss how database security at the application level is incorporated in your project
  - (3 points) — Submit code snippet(s) to illustrate how security aspect is implemented and to support your discussion

- (5 points) Reflection and peer evaluation *(due when the final deliverable is due)*
  - **This is an individual task.** Do **not** include it in the project's final report.
  - Everyone is required to submit the peer evaluation, even though you are working on the project individually.

(155 points) Application:

The following items will be graded during your demonstration. Please have your source code available. You are expected to explain your code snippets and discuss how these requirements are implemented or embedded in your code.

You are required to demo your app **using the program you officially submitted.** Modification after the submission deadline is not allowed and we cannot give you credits.

Since this is a database system project, it is crucial that **your app must successfully connect and interact with the MySQL database** to provide services. If the app cannot successfully connect and/or interact with the MySQL database, **no points** will be awarded (a zero grade will be assigned to the application portion of the final deliverable).

- (10 points) — The app properly utilizes at least 10 different, normalized tables designed in the previous milestones
- (5 points) — The app includes all functionalities and necessary features you proposed to fulfill the project requirements
- (5 points) — Dynamic behaviors are properly included and functioning
- (85 points) — Allow users to perform the following functionalities **through your software**
  - (15 points) Retrieve data from the database
  - (15 points) Add data to the database
  - (15 points) Update data in the database
  - (15 points) Delete data from the database
  - (25 points) Do at least one of the following:

    If your team consists of 5 members, you are required to have **all four** of following functionalities

    - Sort data
    - Search or filter data
    - Export data (from the database) in some forms (such as JSON, XML, HTML, or CSV)
      - Reminder: this must be done through your app. Exporting data directly from the database management system (such as MySQL, Cloud SQL, or phpMyAdmin) does not satisfy this requirement.
    - Upload / import data (in some forms such as JSON, XML, HTML, or CSV) into the database

    - Reminder: this must be done through your app. Importing data directly using the database management system (such as MySQL, Cloud SQL, or phpMyAdmin) does not satisfy this requirement.
- (15 points) — Support multiple users, allowing users to use the app simultaneously without interfering with each other
- (15 points) — Allow returning users to access / retrieve / manipulate their existing data.
- (20 points) — Properly incorporate database security
  - (10 points) Proper database security at the database level
  - (10 points) Proper database security at the application level

The entire app must be functioning properly

What does this mean? All of the following requirements must be satisfied:

- Your app must be a **single unit**. It must not be disconnected. All parts must be properly connected. That is, once a user enters your app, there is an appropriate navigation system and / or an execution flow allowing a user to achieve his / her goal(s) without (re)visiting another part of your app manually (for example, by entering another URL in a web browser's address bar to access another functionality of your app).
- The interactions between your app and the database must be correctly implemented and functioning properly.
- Overall, your entire project must work as expected; it must provide normal behavior, given the domain application of the project.

[10 extra credits] — Host your database and app on GCP

- (5 extra points) — **successfully** host your **database on GCP**
  - Your app must interact and use this database to provide services.
  - All or nothing; no partial credit.
  - No point will be awarded if you do not use the database hosted on GCP when demonstrating your project. To qualify for the extra credits, **your entire demonstration must use the database hosted on GCP.**
  - (-2 points) — Not submit screenshot(s) showing your database on GCP
- (5 extra points) — **successfully** host your **app on GCP**
  - Your **entire app must be functioning properly**
  - All or nothing; no partial credit.
  - No point will be awarded if you do not use the app hosted on GCP when demonstrating your project. To qualify for the extra credits, **your entire demonstration must use the app hosted on GCP.**
  - (-2 points) — Not submit a URL to your app hosted on GCP.

(10 points) User demonstration:

- (5 points) — Application is demonstrated clearly.
- (5 points) — Code segments clearly explained (the graders will ask you to explain your source code which may include the app source code, your SQL statements/commands, and interfacing between your app and your database)

You will sign up to demo your project. An announcement with details will be sent out when a demo sign-up sheet becomes available. **Please only sign up for one slot** and **do not remove the slot of another group.**

Demo sign-up sheet: ***TBD***

A demo is limited to **15 minutes** (10 minutes demo + 5 minutes Q&A). To ensure that all teams are presented in a timely manner, we must ask you to stop when the time limit has been reached. Therefore, we recommend rehearsing the entire presentation and application demo.

To ensure all projects are realistic and demo and grading are completed in a timely manner, the following grading details have been added to the project's final deliverable.

- (-10 points) — Not include **all** necessary files to deploy and run your project
- (up to -50 points) — The project is unrealistic; database design is unreasonable or inappropriate or bad design; or the project and / or database design contain major flaws with respect to database-related concepts, including but not limited to, database design, querying, database interfacing
- (-10 points) — **All partners must be present at the demo.** Otherwise, 10 points will be deducted and you are required to reschedule the demo.
- (-5 points) — More than 5 minutes late for demo
- (-10 points) — Cancel or change the demo time less than 2 hours before the scheduled time
- (-10 points) — No show for the 1st signed up demo
- (-155 points for application, -10 points for user demo, no GCP extra credit) — No show for the 2nd signed up demo
- (-155 points for application, -10 points for user demo, no GCP extra credit) — No demo
- (-100% of the entire final deliverable, no GCP extra credit) — No submission (regardless of whether you attempt to demo/showtime or not)

Personal effort (if you work with a team): **up to 100% penalty** may be deducted based on your contribution and the peer evaluations — **and** you will **not** receive any extra credits — **no GCP extra credit, and no vote winner extra credit**

Brainstorm detailed rubric (https://docs.google.com/document/d/1OqwjsvcdtI8cNPBSJaVi2cADcswvgUhEj1zagsssbaQ/edit?usp=sharing) — Please provide your comment, concern, feedback, question, suggestion by the

due date of the project final deliverable so that the teaching team can review & discuss before finalizing the grading rubric.

As mentioned in class, this is the first time we are implementing this detailed rubric brainstorm, please feel free to suggest how/what we may try to make the brainstorm activity more effective. For any other suggestions, please refer to help yourself learn (../syllabus.html#help-yourself) section of the syllabus or access the *Help yourself learn (https://forms.office.com/r/P20fF1JMnE)* form directly.

Submission

What is due:

Include all team member names and computingIDs in the report. Save your report as a single PDF. ***No Word document. No handwriting. No hand-drawing***.

Some teams may have a very huge zipped file that Gradescope is unable to handle. You may

- **Option 1:** put all your source code file(s) in **GitHub** — do not zip them.
  - When submitting your work to Gradescope, put the link to your GitHub in the report.
  - Be sure to **give proper access** to your repo. If the grader(s) cannot access your source code file(s), we have to assign a zero grade.
- **Option 2:** put all your source code file(s) in **Google Drive** — do not zip them.
  - When submitting your work to Gradescope, put the link to your Google Drive in the report.
  - Be sure to **give proper access** to your shared drive/folder. If the grader(s) cannot access your source code file(s), we have to assign a zero grade.

Include the link to your GitHub repo or Google Drive in your report. Save your report as a PDF file.

Upload your **report (`.pdf`)** to the **Project Final Deliverable assignment on Gradescope**. Make sure you connect your partner to your group on Gradescope so that everyone receives credit.

Each team submits only **one** copy.

Please note, we may randomly deploy and run projects on similar deployment environments. It is important that you submit all the source code and configuration files as well as instructions needed to deploy and run your project.

Making your submission available to the course staff is **your** responsibility; if we cannot access or open your file, we have to assign a zero grade. Be sure to test access to your file before the due date.

Collaborate professionally and contribute fairly

You are expected to **work as a team**. The teaching team will consider **all** reflections and evaluations when assigning the final deliverable grade to an individual team member.

Personal effort (if you work with a team): **up to 100% penalty** — *and* **no GCP extra credit, and no vote winner extra credit** — may be applied based on your contribution and the peer evaluations

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Special Helper Extra Credits

These extra credits are applicable to milestones 1 (DB design) and 2 (DB setup and SQL), and project final deliverable. No special helper extra points for project proposal and showtime.

[Purpose: promote collaboration]
[Total: 10 points per assignment]

If you get help from your classmates (excluding your project partners), when you submit your assignment, please make a note in your report, including the following information:

- Clearly specifies the name(s) and computing id(s) of your special helper(s), and
- Clearly and briefly describe what and how you were helped. If you had multiple helpers, please specify their names and what/how they helped you explicitly. If you do not acknowledge and report your special helper(s), they will not receive extra credit.

Note: Extra points will not be awarded if any of the above information is missing.

The helper earns 1 extra point for each project he/she helps, up to 10 (total) extra points per assignment.

Acknowledgments:
*** This course is generously supported by Google by giving us access to their Cloud Platform (https://cloud.google.com/).*

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Last updated 2026-01-07 18:18